9th Class

CHEMISTRY

Better Notes Brighter Future

Chapter 1

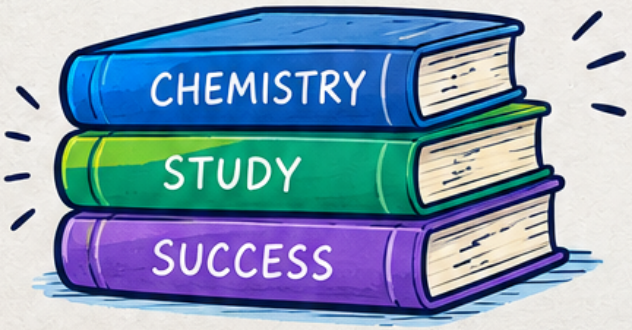
States of Matter and Phase Changes

Complete Handwritten Notes

Learn
Understand
Score High

What's Inside?

- Key Concepts
- Definitions
- Important Points
- Diagrams & Tables
- Examples
- Perfect for Exams



Presented by

Awais

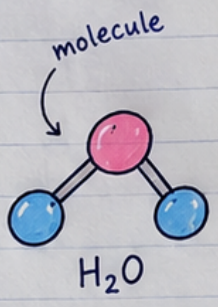
Small Steps Big Results

Chemistry Makes Sense

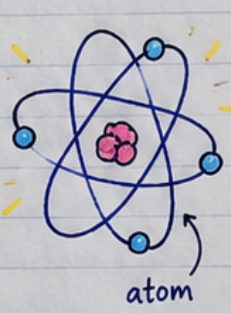
Chapter 1 – States of Matter and Phase Changes

1.1 What is Chemistry?

Chemistry is the branch of science which deals with the properties, composition and structure of substances. It also deals with the physical and chemical changes in matter and the laws or principles which govern these changes.



- "**Composition**" means finding the percentages of elements and compounds in a sample of matter.
- "**Structure of matter**" means the arrangement of atoms in matter.
- **Physical and chemical changes** may be brought about by absorption or evolution of energy.

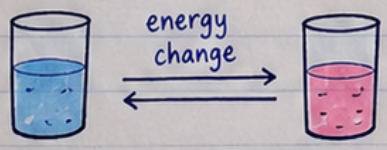


Branches of Chemistry

Chemistry is divided into many branches so that scientists can focus on specific areas and understand this vast subject more easily.

1 Physical Chemistry

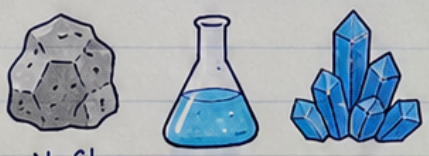
- Studies how substances behave at atomic and molecular levels.
- Explains properties, energy changes and reaction rates.



e.g. study of states of matter, thermodynamics, reaction rates.

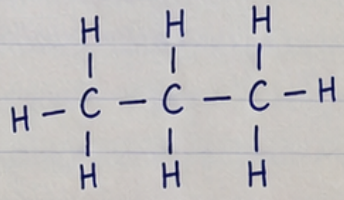
2 Inorganic Chemistry

- Studies the synthesis, composition, properties and structure of substances that contain little or no carbon, such as metals, salts, acids and bases.



3 Organic Chemistry

- Deals with carbon compounds (hydrocarbons and their derivatives), their structure, properties, composition and reactions.



e.g. hydrocarbons

Chapter 1 - States of Matter and Phase Changes

Branches of Chemistry

4 Environmental Chemistry

- Studies chemical and biochemical phenomena in air, soil and water.
- Helps understand pollution and its effects on the environment.
- Develops methods to reduce and remediate environmental pollution.

5 Analytical Chemistry

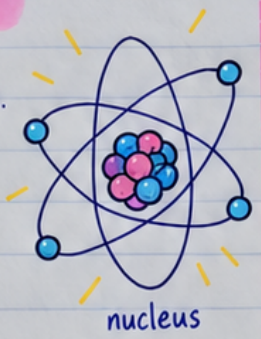
- Deals with the separation, identification and measurement of the concentration of substances in a sample.
- Develops and uses techniques such as chromatography, spectroscopy and titration.

6 Biochemistry

- Studies the chemical substances and vital processes in living organisms.
- Focuses on biomolecules such as proteins, carbohydrates, lipids and nucleic acids.

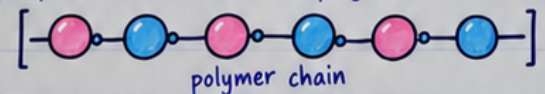
7 Nuclear Chemistry

- Deals with reactions that occur in the nucleus of atoms.
- Studies radioactivity and nuclear transformations such as fission and fusion.
- Has applications in energy production, medicine and radiometric dating.



8 Polymer Chemistry

- Focuses on large molecules (polymers) formed by linking together smaller building blocks (monomers).
- Studies the synthesis, structure, properties and uses of polymers.



9 Geochemistry

- Studies the chemical composition of the Earth, including rocks, minerals, soil, water and the atmosphere.
- Helps explain natural processes such as the rock cycle, formation of minerals and movement of elements in the environment.

10 Medicinal Chemistry

- Designs and synthesizes chemical compounds that can be used as medicines.
- Studies drug action, including delivery, absorption, metabolism and how drugs interact with the body.

11 Astrochemistry

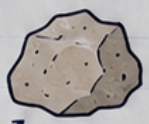
- Studies molecules and ions in space, including in stars, planets, comets and interstellar clouds.
- Investigates how these particles form and interact with radiation in extreme conditions.

Chapter 1 – States of Matter and Phase Changes

1.2 States of Matter

1 What is Matter?

- Matter is anything that **has mass and occupies space** (has volume).
- Everything around us – the Earth, air, water, living things – is made up of matter.
- The world is made up of both **matter** and **energy**. Energy is **non-material** – it does not have mass or take up space.



matter (has mass and occupies space)

energy (non-material, no mass, no volume)

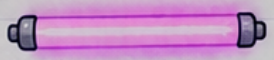
2 The Three Primary States of Matter

- **Solid:** definite shape and volume, particles closely packed in a regular arrangement, strong intermolecular forces, not easily compressed, particles vibrate about fixed positions.
- **Liquid:** definite volume but no fixed shape (takes shape of container), particles close together but disordered, moderate forces, not easily compressed, particles slide past each other.
- **Gas:** no fixed shape and no fixed volume (fills container), particles far apart, very weak forces, easily compressed, particles move rapidly and randomly.

3 Plasma – A High-Energy State

- Plasma is a high-energy state of matter.
- It is found in fluorescent tubes, lightning and welding arcs.
- Plasma is a partially ionized gas containing electrons, positive ions and photons (light energy).
- It has high temperatures and conducts electricity.

lightning (plasma)

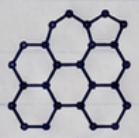


fluorescent tube (plasma)

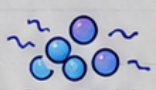
4 Intermediate and Exotic States

- **Supercritical fluids:** occur at high temperatures and pressures; have properties of both liquids and gases (e.g. supercritical CO_2 used in decaffeination).
- **Liquid crystals:** have properties between liquids and solids; used in LCD screens.
- **Graphene:** a single layer of carbon atoms with unique properties (an exotic form of solid matter).
- **Dark matter:** a proposed form of matter that does not emit, absorb or reflect light.
- **Bose-Einstein condensate:** occurs at temperatures close to absolute zero, where atoms behave as a single quantum system.

liquid crystal screen (LCD)



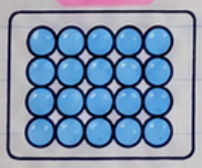
graphene (carbon atoms)



Bose-Einstein condensate

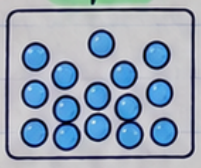
Particle Diagrams

Solid



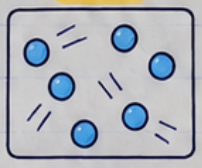
- closely packed
- regular arrangement
- vibrate in place

Liquid



- close together
- disordered
- slide past each other

Gas



- far apart
- random arrangement
- move rapidly

Comparison of the Three States

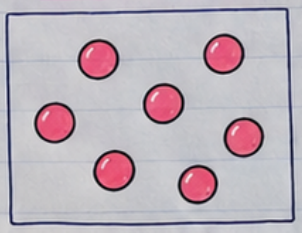
Property	Solid	Liquid	Gas
Shape	fixed	no fixed shape	no fixed shape
Volume	fixed	fixed	no fixed volume
Particle arrangement	close & regular	close & disordered	far apart & random
Intermolecular forces	strong	moderate	very weak
Compressibility	not easily compressed	not easily compressed	easily compressed
Particle movement	vibrate in place	slide past each other	rapid random motion

1.3 Element, Compound and Mixture

1 Element

- The simplest form of matter.
- A pure substance containing only the same kind of atoms.
- Cannot be broken down into simpler substances by ordinary chemical reactions.
- May exist as a solid, liquid or gas.
- Examples: oxygen (O_2), copper (Cu), sulfur (S).

Element: oxygen



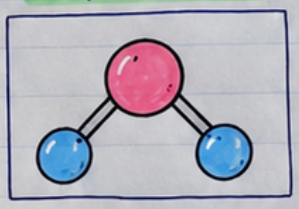
Oxygen (O)

Each particle represents an oxygen atom. All particles are the same.

2 Compound

- A pure substance made from two or more different elements chemically combined in a fixed ratio.
- When elements come together, they form chemical bonds that are not easy to break.
- Compounds can be molecular (made of non-metals) or ionic (made of metals and non-metals).
- Examples: water (H_2O), ammonia (NH_3), sodium chloride (NaCl).

Compound: water



H_2O

Different kinds of atoms are chemically bonded together in a fixed ratio.

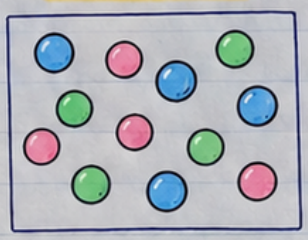
= oxygen atom (O)

= hydrogen atom (H)

3 Mixture

- Contains more than one type of element or compound mixed together in any ratio.
- The components are not chemically bonded.
- Each substance keeps its own properties.
- Examples: air, soil, milk, tap water.

Mixture: air



Different particles are mixed together. They are not chemically bonded and can be separated by physical methods.

Types of mixtures

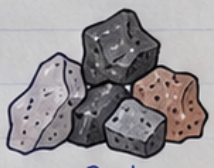
Homogeneous mixture

- Components are evenly mixed throughout.
- Same composition in all parts.
- Cannot see the different components.
- Example: salt solution (salt dissolved in water).

Salt solution (homogeneous)

Heterogeneous mixture

- Components are not evenly mixed.
- Different composition in different parts.
- Can see the different components.
- Example: rock (different minerals).



Rock (heterogeneous)

Comparison

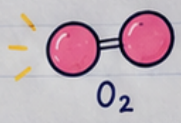
Feature	Element	Compound	Mixture
Composition	Same kind of atoms	Two or more elements chemically combined (fixed ratio)	Two or more substances mixed (any ratio)
Chemical bonds	No (cannot be broken down by ordinary reactions)	Yes (atoms are chemically bonded)	No (components not bonded)
Can be separated?	No	No (only by chemical reactions)	Yes (by physical methods)
Examples	Oxygen, copper, sulfur	Water, ammonia, sodium chloride	Air, soil, milk, tap water

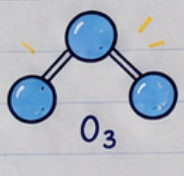
1.4 Allotropic Forms of Substances

Some elements can exist in more than one structural form with different physical and chemical properties. These different forms are called **allotropes**, and the phenomenon is called **allotropy**.

same element, different structure!

Allotropes of Oxygen





- Oxygen exists as two allotropes: oxygen (O_2) and ozone (O_3).
- O_2 is a colourless, odourless gas, relatively unreactive and essential for respiration.
- O_3 (ozone) is a pale blue gas, more reactive and a stronger oxidising agent.

Allotropes of Carbon

Carbon has several allotropes with very different structures and properties.

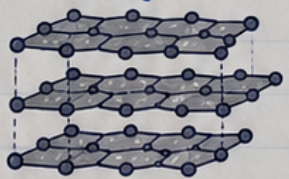
1 Diamond

- Giant macromolecular structure.
- Each carbon atom is covalently bonded to 4 other carbon atoms in a tetrahedral arrangement.
- Very hard, high melting point, does not conduct electricity.

diamond crystal (tetrahedral structure)

2 Graphite

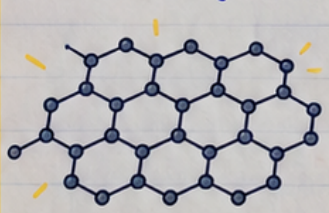
- Layers of carbon atoms arranged in hexagonal rings.
- Layers are held together by weak forces, so they can slide over each other.
- Soft, slippery and does conduct electricity.



layers of hexagonal carbon rings

3 Graphene

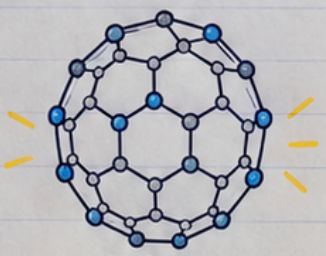
- A single layer of carbon atoms in a hexagonal pattern.
- Very tough, flexible and light.
- High resistance (strength) and good conductor of electricity.



single layer of carbon atoms (hexagonal pattern)

4 Buckminster Fullerene

- C_{60}, a spherical cage-like structure made of 12 pentagons and 20 hexagons.
- Each carbon atom is bonded to 3 others.
- Soft and does not conduct electricity.



C_{60} (Buckminster fullerene)

Allotropes of Sulfur

- Sulfur exists in at least two allotropes: rhombic sulfur and monoclinic sulfur.
- They have the same chemical composition but different crystal structures and physical properties. Rhombic sulfur is more stable at room temperature.

Rhombic Sulfur

- More stable form.
- Yellow, octahedral crystals.
- Melting point: 113 °C.

rhombic sulfur (octahedral crystals)

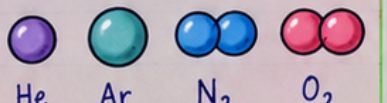
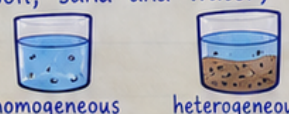
Monoclinic Sulfur

- Less stable form (converts to rhombic sulfur on standing).
- Yellow, needle-shaped crystals.
- Melting point: 119 °C.

monoclinic sulfur (needle-shaped crystals)

Chapter 1 – States of Matter and Phase Changes

1.5 Differences between Elements, Compounds and Mixtures

Feature	Element	Compound	Mixture
Definition	The simplest form of matter. A pure substance made up of only one type of atom.	A pure substance formed by the chemical combination of two or more elements in a fixed ratio.	An impure sample containing more than one type of element or compound, not chemically combined.
Purity	Pure substance.	Pure substance.	Impure sample.
Ratio of components	Only one type of atom.	Fixed ratio of elements (e.g. H_2O always 2:1).	Components can be in any ratio.
Can it be broken down chemically?	No. It cannot be broken down into simpler substances by chemical means.	Yes. It can be broken down into its constituent elements by a chemical reaction.	No (by chemical means), but it can be separated into its components by physical methods.
Do components retain their own properties?	Not applicable (only one type of atom).	No. The elements lose their original properties and form a new substance with different properties.	Yes. Each component retains its own properties.
How it is represented	By a chemical symbol (e.g. He, Fe) or a molecular formula (for molecular elements).	By a chemical formula (e.g. H_2O, NH_3, HCl).	Not by a fixed formula. Often described by the names of its components (e.g. air, saltwater).
Examples	He, Ar (monoatomic) N_2, O_2, Cl_2 (molecular) 	H_2O, NH_3, HCl	Homogeneous or heterogeneous mixtures (e.g. air, saltwater, soil, sand and water) 

Key Points to Remember

- Elements are pure and cannot be broken aown chemically.
- Compounds are pure and have a fixed ratio of elements.
- Mixtures are impure and can be separated by physical methods.
- Mixtures can be homogeneous (uniform) or heterogeneous (non-uniform).

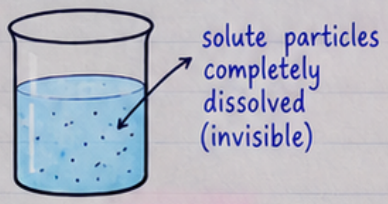
Exercise Corner

1. Which elements are found in pure state on Earth?
2. Which elements are present in very small amounts on Earth?

1.6 Solution, Colloidal Solution and Suspension

True Solution

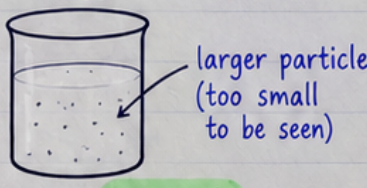
- Solute particles are completely homogenized in the solvent.
- Particles are very small and not visible.
- Passes through filter paper.
- Example: sodium chloride solution (salt water) or copper sulphate solution.



Solution
(e.g. NaCl solution)

Colloidal Solution

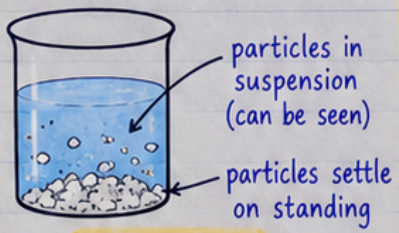
- Particles are larger than in a true solution but still very small and not visible to the naked eye.
- Do not settle on standing (i.e. remain evenly distributed).
- Passes through filter paper.
- Example: starch solution or egg white.



Colloid
(e.g. starch solution)

Suspension

- Particles do not dissolve in the solvent.
- Particles are large enough to be seen.
- Settle on standing.
- Can be separated by filtration (do not pass through filter paper).
- Example: chalk in water.



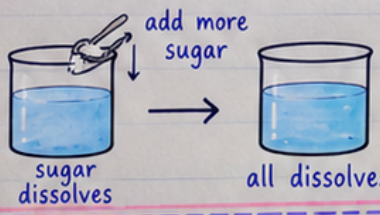
Suspension
(e.g. chalk in water)

1.7 Unsaturated and Saturated Solutions

A solution is made by dissolving a solute in a solvent. The amount of solute that can dissolve depends on the temperature and the type of solute.

Unsaturated Solution

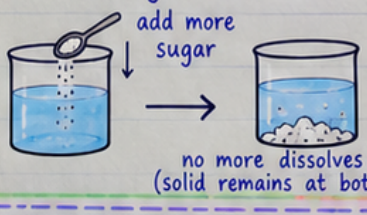
- As long as more solute can dissolve in a solvent at a given temperature, the solution is unsaturated.
- In the sugar-in-water experiment, adding more sugar still dissolves and no solid remains at the bottom.



Unsaturated solution (more solute can dissolve)

Saturated Solution

- When no more solute can dissolve in a solvent at a given temperature, the solution is saturated.
- In the sugar-in-water experiment, further addition of sugar does not dissolve and the extra sugar remains at the bottom.



Saturated solution (no more solute can dissolve)

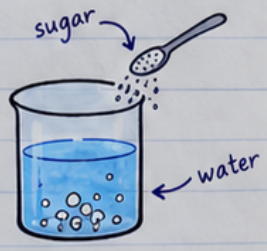
Different solubilities

- Different solutes have different solubilities in a solvent.
- At 20°C, sodium chloride has a solubility of about 36 g per 100 g of water, while table sugar (sucrose) has a solubility of about 200 g per 100 g of water.

1.8 Effect of Temperature on the Solubility of Solutes

What is Solubility?

Solubility is the amount of solute which can dissolve in 100 g of a solvent at a particular temperature.



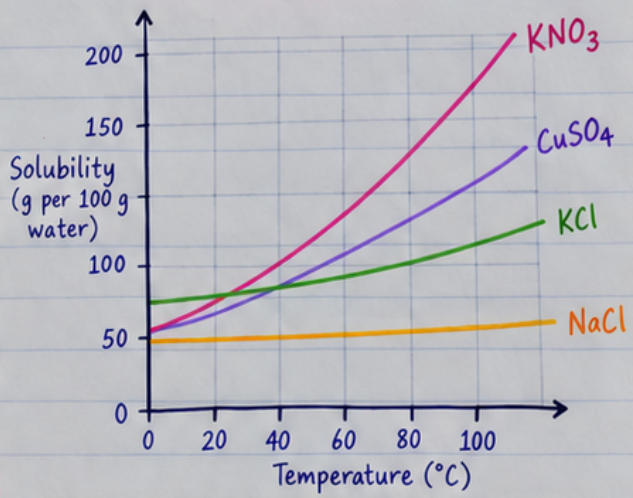
How Temperature Affects Solubility

- In general, the solubility of most solid solutes increases with temperature, but this is not always the case.

Solubility Increases with Temperature ↑

- Many ionic compounds become more soluble as temperature increases, for example: potassium nitrate (KNO_3), silver nitrate ($AgNO_3$), potassium chloride (KCl), copper sulphate ($CuSO_4$) and sodium nitrate ($NaNO_3$).
- These compounds show a large increase in solubility at higher temperatures.
- Sodium chloride ($NaCl$) changes very little with temperature (almost flat).

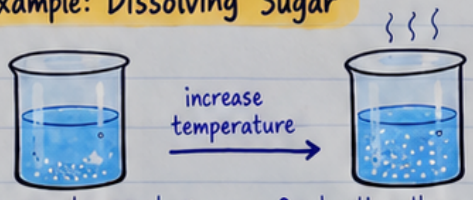
Solubility Curves



Solubility Decreases with Temperature ↓

- Some compounds become less soluble as the temperature increases, for example: lithium carbonate (Li_2CO_3), calcium chromate ($CaCrO_4$) and calcium hydroxide ($Ca(OH)_2$).
- The solubility of gases in water also decreases as temperature increases (e.g. oxygen, carbon dioxide), so warm water holds less dissolved gas than cold water.

Example: Dissolving Sugar



At room temperature (20 °C) e.g. 20 g of sugar dissolves.

On heating the water (60 °C) e.g. 40 g of sugar dissolves.

More sugar dissolves at higher temperature.

Key Points – Chapter 1 Recap

- Chemistry is the branch of science that studies the properties, composition and structure of substances.
- Chemistry has three main branches: physical, inorganic and organic chemistry.
- The three common states of matter are solid, liquid and gas. Plasma is another state and there are intermediate states such as liquid crystals.
- Matter can be elements, compounds or mixtures.
- Solutions are homogeneous mixtures and colloids are mixtures with larger dispersed particles.
- The solubility of most solid solutes increases with temperature (e.g. KNO_3, $AgNO_3$, KCl, $CuSO_4$, $NaNO_3$).
- Sodium chloride changes very little with temperature.
- Some solids (e.g. Li_2CO_3, $CaCrO_4$, $Ca(OH)_2$) and gases become less soluble as temperature increases.

Revision Tip!

Try drawing the solubility curves from memory and label a few familiar compounds!